ATI Water Monitor App: User Manual

Entry, automatic calculation, shared Google Sheet and comparison with other ATIs

1. What the app does

ATI Water Monitor is a smartphone app for recording the water level of the FP (Farmers' Practice) plot and the 3F4D+MD plot. You can enter data without internet. The numbers you enter are pasted into a shared Google Sheet prepared by the teacher, and you can compare your ATI with the other ATIs in graphs. No script and no password are needed.

Step	What happens
1. Enter	Students enter water level, irrigation, rain, fertilizer, pesticide and tiller counts in the app. Water level, irrigation and so on are calculated automatically.
2. Paste	Press "Copy my data" and paste the numbers below the last row of the tab "Data" in the shared sheet.
3. Compare	Enter the "compare link" in the Compare tab and load it: your ATI is shown against the other ATIs in the sheet.

2. Teacher setup (shared Google Sheet)

The administrator (for example a teacher) does this once. You do not need one sheet per ATI: all ATIs that want to compare use the same sheet.

1. Upload the template Excel file ATI_Monitor_Shared_TEMPLATE.xlsx to Google Drive, open it and choose "File > Save as Google Sheets".
2. **Sharing link (for entering)**: click "Share", set "Anyone with the link" to "Editor", and copy the link. Students paste their data here.
3. **Compare link (for reading)**: open "File > Share > Publish to web". Under "Link", choose the sheet "Data" instead of "Entire document", choose the format "Comma-separated values (.csv)" and press "Publish". Copy the link shown.
4. Give the students three things: (1) the app address, (2) the sharing link, (3) the compare link.

> **Note:** Anyone who knows the sharing link can edit or delete the sheet. If rows are deleted by mistake, restore them with "File > Version history". Anyone who knows the compare link can read the data (water level, irrigation and so on; no farmer names). To stop publishing, press "Stop publishing" on the "Publish to web" screen.

3. How students use it

3.1 The first time

- Open the app. On a phone, open it in Chrome and choose "Add to Home screen" to start it from an icon.
- **Setup** tab: enter ATI name*, place*, sowing date*, seedling age at transplanting*, and TP and b* of FP and 3F4D+MD (* = required). While a required item is empty, "Save entry" in the Entry tab cannot be pressed.
- **Schedule** tab: planned dates appear automatically. Enter the date you actually did the work in "Actual".

- **Share** tab > Shared Google Sheet: type the sharing link and the compare link.

How to read m, TP and b: see the figure below. m = tape reading at the water surface, TP = tape reading at the top of the pipe, b = height of the pipe top above the ground (measured with another scale). Water level = m − TP + b. A negative value means the water is below the ground.

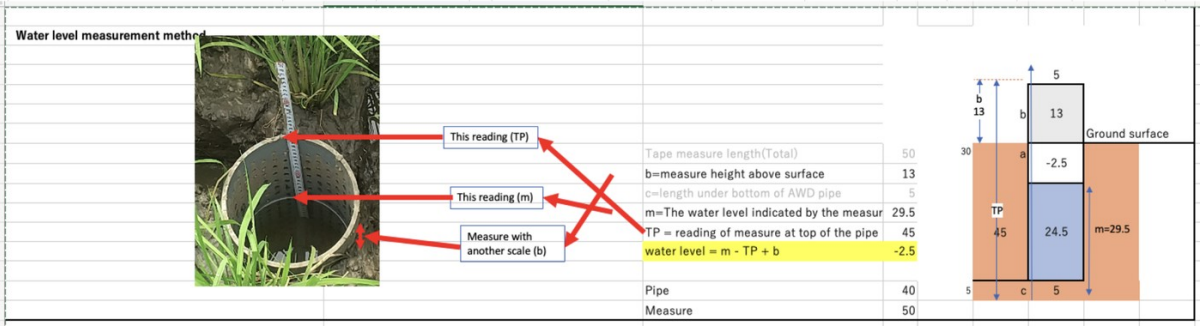


Figure: how to measure the water level (example: m = 29.5, TP = 45, b = 13 gives a water level of −2.5 cm)

3.2 Every day

1. **Entry** tab: choose the date and enter the tape reading (m) of FP and 3F4D+MD. The water level is calculated for you. If you irrigated, choose "Yes" and enter the readings before and after irrigation.
2. "Save entry" becomes active when all required items (*) are filled and the values are in range. Missing items are shown in red under the button.
3. **Share > Copy my data**. Open the shared sheet with the "Open shared sheet" button, click the first empty cell of column A in the tab "Data" (below the last row) and paste (Ctrl+V; on a phone, long-press and Paste). Do this when you are online.

Input	Required items
Setup	ATI name, place, sowing date, seedling age, TP and b (both plots)
Entry (water level)	Date, m of FP, m of 3F4D+MD. If irrigation is "Yes": the readings before and after irrigation of that plot
Field records: fertilizer	Date, plot, at least one fertilizer amount (kg)
Field records: pesticide	Date, plot, type, chemical name
Field records: tillers	Date, plot, tiller counts of all 5 hills (whole numbers). The value used is the mean of the 2nd, 3rd and 4th largest
Re-measure TP and b	Plot, date, TP, b

3.3 Correcting entries

- Water-level entry: press ✎ in the Entries list, change the values and press "Save entry" again (the same date is overwritten). ✕ deletes the day.
- Fertilizer, pesticide, tillers, TP and b: press ✎ in the list, change the values and press "Update". Press "Cancel" to stop.
- After correcting, press "Copy my data" again and paste below the last row. Only the newest paste of the same ATI is used for comparison (older pastes are ignored).

4. Comparing with other ATIs (Compare tab)

- Enter the "compare link" in the Share tab, then press "Load from shared sheet" in the Compare tab to read the other ATIs. Tick the ATIs you want to see.
- Your own ATI is drawn with thick lines. The x axis is days after transplanting (DAT).
- Graphs: total irrigation and saving (FP vs 3F4D+MD), cumulative irrigation, water level, tiller number.
- "View" switches the display. "Overlay" draws all ATIs in the same graph. "Each ATI" shows a separate panel with water-level and cumulative-irrigation graphs for every ATI. It also works without data of your own (for example when a teacher looks at all ATIs).
- To see how Compare looks without internet, use "Try with demo data" in Setup (demo ATIs also appear in Compare).

5. Demo, backup and other points

- **Demo:** "Try with demo data" in Setup opens the app with example data. Nothing is saved. You can try Compare too. Press "Exit demo" at the top to leave.
- **Backup:** Data are stored only in the browser of that phone. Use "Export file (JSON)" in Share from time to time. Use it also to move to another phone.
- **View-only link:** "Create link" in Share also makes a link that shows only the dashboard, without the sheet.
- **Language:** The button at the top right switches English and Bengali. The Bengali is a draft and needs checking.

6. Notes and limits

- Each ATI name must be unique. Pastes with the same name are treated as the same ATI.
- Each paste contains all data of that ATI (not a difference), so the sheet grows with every paste. The teacher may delete old pastes from time to time (rows whose "Upload" in column B is older). One paste is about 190 rows.
- Column B "Upload" is a number for the date and time of the paste (year, month, day, hour, minute). The app fills it in. Do not change it, the header row, or the order of the columns.
- The app can read dates even if the sheet converts them to another format.
- Verified: the app's calculations equal the original Excel values; required-item checks, corrections, demo, the format of the copied rows and loading (with mock data in place of Google).
- Not verified: pasting into a real Google Sheet and loading the published CSV, behaviour on real phones, correctness of the Bengali. Test with the first ATI before handing it out.